

Exercise 7

1. Execute Bisecting K-Means with $K = 3$ leaf clusters using Euclidean distance on the data below: assign all objects to a cluster, compute the SSE and find the centroid positions in each iteration. Instead of a random initialization the initial centroids are p_{10} and p_{11} .

ID	p1	p2	p3	p4	p5	p6	p7	p8	p9	p10	p11
x	1.0	2.0	3.0	3.0	2.0	2.0	2.5	4.0	6.0	7.0	6.5
y	1.0	2.0	1.0	2.5	8.0	6.0	7.0	7.0	2.0	3.0	1.0

2. Explain the following concepts used by the DBSCAN algorithm:
 - hyperparameters $minPts$ and ε
 - core points
 - border points
 - noise points
3. Explain the following concepts of reachability in DBSCAN:
 - directly density-reachable data point
 - density-reachable data point
 - density-connected data points
4. Explain how the DBSCAN algorithm works. How does it find clusters after assigning points to core, border and noise points?
5. Cluster the points below using DBSCAN with $\varepsilon = 1.5$ and $minPts = 3$. For each point determine whether it is a core, border, or noise point. What are the resulting clusters?

	x	y
A	1.0	1.0
B	1.5	1.5
C	5.0	5.0
D	3.0	4.0
E	4.0	4.0
F	3.0	3.5

